

FEB12001X – Resit 8 July 2025 (online proctored)

FEB12001X Applied Microeconomics 24/25 (ESE) · 7 exercises · 52.0 points

Question 1

10 pts · Last updated 30 June, 08:10 · Saved in Templates ESE

Consider a town with 5000 identical citizens. Each citizen owns a bike and a car. Each citizen must decide whether to travel to work by bike or by car. Travelling by bike would take $m_f = 40$ minutes. Travel time by car depends on whether there are traffic jams. A traffic jam arises when more than 500 citizens take the car. Hence, travelling by car would take $m_c = 25$ minutes if the number of cars $C \leq 500$. Travelling by car would take $m_c = 25 + \frac{1}{100} (C - 500)$ minutes if $C > 500$. Assume that each citizen wants to minimise own travel time and that every citizen decides independently whether to travel by bike or by car.

Text

Determine the number of citizens that travel by car in the Nash equilibrium.

2 pts · Open

+1 point

In the Nash equilibrium, citizens travel by car until travel time by car equals travel time by bike.

+1 point

This gives $40 = 25 + \frac{1}{100} (C - 500)$, which can be solved for $C = 2000$.

Prove that in the Nash equilibrium, inefficiently many citizens travel by car. Explain why this happens.

4 pts · Open

+1 point

The socially efficient number of cars can be found by minimizing total travel time

+1 point

$\min_C (5000 - C)40 + C \left(25 + \frac{1}{100} (C - 500) \right)$, which yields first-order condition $\frac{1}{50} C - 20 = 0$, which can be solved for $C^* = 1000$.

+2 points

This is due to the Tragedy of the Commons / a negative externality: each individual who chooses car does not take into account that this implies that other car users spend more time in the traffic jam.

Two politicians debate on possible improvements. The first politician argues that traffic jams can be reduced by investing in road capacity. This politician proposes to increase the number of car lanes on the road, such that traffic jams only arise if there are more than 2000 cars on the road. After this investment, travel time by car would be $m_c = 25$ minutes if the number of cars $C \leq 2000$ and $m_c = 25 + \frac{1}{100} (C - 2000)$ if $C > 2000$.

The second politician argues that it would be better to invest in better bike lanes instead. At similar investment cost, it would be possible to reduce travel time by bike to $m_f = 35$ minutes.

Text

Calculate for both of these proposals the reduction in total travel time compared to the situation in question a).

2 pts · Open

+1 point

Proposal extra car lanes: New Nash eq: $40 = 25 + \frac{1}{100}(C - 2000)$ such that $C = 3500$. Travel time: still 40 minutes for everyone, so no reduction in travel time.

+1 point

Proposal extra bike lanes: New Nash eq: $35 = 25 + \frac{1}{100}(C - 500)$ such that $C = 1500$. Everyone now has travel time of 35 minutes. Total reduction in travel time: $5000 * (40-35) = 25000$ minutes.

Other types of government interventions are possible. Consider the choice between a Pigouvian subsidy for citizens who travel by bike and a Pigouvian tax for citizens who travel by car.

Text

Suppose the government only cares about efficiency in citizens' choice between travel by bike or by car. Discuss which of these two interventions is better and why. Use maximally three sentences.

2 pts · Open

+2 points

Both types of intervention yields the same outcome: citizens' choice between bike and car is socially efficient. This holds by definition of Pigouvian subsidies and taxes.

Another way of arguing is as follows. For efficiency, it only matters how many citizens choose car versus bike. As this is a binary choice, the relative cost and benefit of each choice matters, such that it does not matter whether choice bike becomes more attractive or choice car less attractive.

Question 2

6 pts · Last updated 30 June, 08:10 · Saved in Templates ESE

Consider the market for second-hand units of good X . Suppose that the quality of a second-hand unit of good X is either high or low. Fraction λ of these units is high-quality, where $0 < \lambda < 1$ such that fraction $1 - \lambda$ is low-quality. There are many potential buyers and sellers of second-hand units of good X who all may want to buy or sell one unit. Each seller knows the quality of the unit they sell. A seller sells when the price is at least as high as their valuation. If the unit has low quality, the seller's valuation of the unit equals $S_L = 100$ euro. If the unit has high quality, the seller's valuation equals $S_H = 500$ euro. Buyers do not know the quality of a given unit of good X (but do know that a fraction λ of all units has high quality). Buyers' valuation of a low quality unit is $B_L = 300$ euro and of a high quality unit is B_H euro. Assume that $B_H > 700$. A buyer buys if their expected valuation of the unit is at least as high as the price.

In this model, adverse selection arises if $B_H < 300 + \frac{200}{\lambda}$.

Text

Prove that adverse selection arises if $B_H < 300 + \frac{200}{\lambda}$.

2 pts · Open

+1 point

Buyers are maximally willing to pay their expected valuation. Adverse selection arises if what buyers are willing to pay is below what **high-quality** sellers want to get.

+1 point

Buyers are maximally willing to pay $(1 - \lambda) 300 + \lambda B_H$. A seller of a high-quality unit is willing to sell for a price of minimally 500 euro. Thus adverse selection arises if $(1 - \lambda) 300 + \lambda B_H < 500$.

This can be rewritten in steps to $\lambda (B_H - 300) < 200$ and to $B_H < 300 + \frac{200}{\lambda}$.

The condition in the previous question shows that if λ is larger, the (maximum) level of B_H at which adverse selection arises is smaller. Explain why in at most three sentences.

2 pts · Open

+2 points

Higher λ implies that for buyers the probability that a given unit is high-quality is higher. This increases buyers' willingness to pay. As a result, to meet high-type sellers minimum price, buyer's valuation of high-quality items can be lower.

Another way of phrasing is as follows: buyers fear to pay a lot (too much) for a low-quality item. If λ is low, they are only willing to run that risk if their valuation B_H is very high. If λ is higher, they are willing to run that risk for lower values of B_H .

Suppose adverse selection arises in equilibrium. What is the range of prices that can possibly arise on the market for second-hand units of good X ? Explain in maximally two sentences.

2 pts · Open

+2 points

Due to adverse selection, everyone knows that only low-equality items are traded. These transactions can only take place if $S_L \leq p \leq B_L$, which implies that the price p must be between 100 and 300.

Question 3

10 pts · Last updated 30 June, 08:10 · Saved in Templates ESE

The government of an island considers to invest in protection against water by constructing dykes. This costs 1200 euro. There are 60 citizens on the island. For 40 citizens, the benefit of the dykes is equal to 15 euro per citizen. For 20 citizens, the benefit of the dykes is 40 euro per citizen. The construction cost of the dykes are financed by one new tax. The government considers several different taxes.

Text

Suppose the government introduces a lump-sum tax such that every citizen pays the same amount in taxes. Determine whether the construction of the dykes is a Pareto-improvement.

2 pts · Open

+1 point

Lumpsum tax implies that per citizen the tax would be $1200/60 = 20$ euro.

+1 point

This means that the citizens with benefit of 15 euro would be worse off: no Pareto-improvement.

Suppose the government would introduce a tax on the consumption of good X . Supply of good X is perfectly elastic and is given by $P = 50$, where P is the price per unit of good X (in euro). Demand for good X of all citizens together is given by $P = 120 - \frac{1}{4}Q$, where Q is quantity demanded of good X . Suppose that the government introduces a unit tax of 5 euro on good X .

Text

Determine the Excess Burden of this tax on good X .

2 pts · Open

+1 point

Equilibrium before tax at $Q = 280$. After tax at $Q = 260$.

+1 point

Excess Burden equals $\frac{1}{2} \cdot 5 \cdot (280 - 260) = 50$.

Now suppose that the government would introduce an ad valorem tax on both good X and on another good Y . Supply of good Y is also perfectly elastic. In the equilibrium without a tax on good Y , the elasticity of demand for good Y is equal to 2. The government aims to keep the total Excess Burden of taxation as small as possible.

Text

Determine whether the tax rate on good X should be lower or higher than the tax rate on good Y .

2 pts · Open

+1 point

Demand elasticity for good X is $\varepsilon_D^X = \frac{1}{\frac{1}{4}} \frac{50}{280} = \frac{5}{7}$.

+1 point

This is smaller than the elasticity of good Y . Using the inverse elasticity rule (lecture 6): tax rate on good X should be higher than the tax rate in good Y .

According to the Ramsey rule (or inverse elasticity rule), in order to minimise the total distortion caused by taxation, goods with high elasticity of demand should be taxed at lower rates than goods with low elasticity of demand. Explain why in maximally 4 sentences.

2 pts · Open

+2 points

For goods with high demand elasticity, consumers have good alternatives for consuming this good, unlike goods with low demand elasticity. As a result, a tax will lead to a greater decrease in quantity demanded (or: a greater substitution effect), which means that the loss in efficiency is relatively high. In order to minimize the total efficiency loss, it is optimal to set a relatively low tax on goods with relatively high demand elasticity.

Assume that the 40 citizens who derive 15 euro benefit from the construction of the dykes are relatively poor, whereas the 20 citizens who derive 40 euro benefits from the construction of the dykes are relatively rich. The marginal utility of income of each of the 20 rich citizens is equal to 1. The marginal utility of income of each of the 40 poor citizens is $\alpha \geq 1$. Furthermore, assume that the government maximises a social welfare function. Social welfare is equal to the sum of the individual utilities of all citizens on the island.

Text

Suppose the government would use the lump-sum tax if it decides to construct the dykes, such that every citizen pays the same amount in taxes. For what levels of α would the government optimally decide to construct the dykes?

2 pts · Open

+1 point

In terms of social welfare, the benefit of the dykes is now $40 \cdot 15\alpha + 20 \cdot 40$. The cost of the lump-sum tax are $40 \cdot 20\alpha + 20 \cdot 20$.

+1 point

Benefits are larger than cost if $40 \cdot 15\alpha + 20 \cdot 40 > 40 \cdot 20\alpha + 20 \cdot 20$ which can be solved for $\alpha < 2$.

Question 4

10 pts · Last updated 30 June, 08:10 · Saved in Templates ESE

A profit-maximizing tobacco company hires a worker. The worker's utility function is:

$U = w - \frac{1}{2}\theta e^2 - \beta q$, where w is the worker's income, e is the amount of effort exerted by the worker, and q is the amount of output produced by the worker. θ and β are positive parameters. The worker's income consists of a base salary a and a bonus b per unit of production. Hence, $w = a + bq$. There are no restrictions on a and b , they can take any value, also negative values. The production function is $q = \mu e$. The first term of the utility function reflects that the worker enjoys income. The second term reflects worker's cost of effort. And the third term reflects that the worker feels guilt about producing tobacco, which reduces his utility. When the worker decides not to work for this company, his utility is equal to V . The worker chooses to work for the tobacco company if his utility is at least equal to V . Effort e cannot be negative.

Text

Suppose the worker is hired by the tobacco company. Derive the condition under which the worker chooses to exert positive effort. Explain the condition in one clear sentence.

3 pts · Open

+1 point

Substitute the income equation and production function into the utility function:

$U = a + b\mu e - \frac{1}{2}\theta e^2 - \beta\mu e$. To find optimal effort, differentiate and set equal to zero:
 $b\mu - \theta e - \beta\mu = 0$.

+1 point

Solve for e , which results in $e = \frac{\mu}{\theta}(b - \beta)$. Hence, effort is positive if $b > \beta$.

+1 point

The worker chooses positive effort when the monetary gain from doing so exceeds the cost of feeling guilty.

Derive a condition describing the lowest level of the base salary a that is needed to induce the worker to choose to work for the tobacco company.

2 pts · Open

+1 point

$U \geq V$ and so $a + b\mu e - \frac{1}{2}\theta e^2 - \beta\mu e \geq V$

+1 point

Rearranging gives $a \geq V + \frac{1}{2}\theta e^2 + (\beta - b)\mu e$. (The lowest level is where this condition holds with equality. It is nice but not necessary to mention this to get the point).

Assume each unit of production is sold by the tobacco company for price p , which is assumed sufficiently high so that the tobacco company wishes to produce a positive amount. Derive the profit-maximizing level of the bonus b .

3 pts · Open

+1 point

Profits equal $pq - a - bq$.

+1 point

Substitute the level of effort chosen by the worker as derived under part 4a and the level of base salary needed to attract the worker derived under part 4b. This results in:

$$(p - \beta) \mu \left(\frac{\mu}{\theta} (b - \beta) \right) - V - \frac{1}{2} \theta \left(\frac{\mu}{\theta} (b - \beta) \right)^2$$

+1 point

Take the derivative to b and set equal to zero: $(p - b) \mu \frac{\mu}{\theta} - \theta \left(\frac{\mu^2}{\theta^2} (b - \beta) \right) = 0$. Solving gives $b = p$. (To be sure: just writing $b = p$ without a derivation does not result in any point.)

Suppose the tobacco company could reduce the level of β , for instance through a marketing campaign targeted at its worker. Give two reasons for why this might increase the company's profits according to the model described above. You do not need to provide any calculations. Do not use more than two sentences.

2 pts · Open

+1 point

A reduction in β allows the company to offer a lower income to the worker. Hence, labor costs are lower.

+1 point

For any wage contract, a decrease in β implies that the worker exerts more effort, resulting in higher production and thus revenues for the company.

Question 5

6 pts · Last updated 30 June, 08:10 · Saved in Templates ESE

A worker has the following utility function: $U = V(a + be) - \delta e$, where a is the base salary, e is worker's effort, and b is a bonus per unit of effort. The function $V(\cdot)$ describes the worker's utility from income. The function is increasing and concave, i.e. $V'(\cdot) > 0$ and $V''(\cdot) < 0$. The last term of the utility function reflects the cost of exerting effort. We assume $\delta > 0$.

Text

Derive the first-order condition describing the level of effort that results in maximum utility.

2 pts · Open

+2 points

$$bV'(a + be) - \delta = 0.$$

Suppose the base salary a increases. What happens to the worker's choice of effort e ? Use the first-order condition to derive your result.

2 pts · Open

+2 points

Effort decreases because an increase in a reduces the marginal utility from income $V'(\cdot)$ and so reduces the marginal benefit from exerting effort (the first term in the first-order condition).

Why might it be reasonable to assume that the utility derived from income is concave rather than linear in income? Explain in one clear sentence.

2 pts · Open

+2 points

An increase in income typically brings more additional utility when income is lower, because with a low income some important needs are not yet satisfied.

Question 6

6 pts · Last updated 30 June, 08:10

A profit-maximizing manager is thinking about prohibiting workers to use social media during work time. Using social media yields a utility gain expressed in euros to the workers of V per worker. Using social media yields a production value loss expressed in euros to the manager of X per worker. Assume that it is costless for the manager to monitor whether workers use social media, and that the manager can impose a non-monetary penalty on workers who do so, which is costless for the manager, but yields a utility loss of F to the worker. We assume that $F > V$.

Text

Under what condition is it socially efficient to prohibit workers to use social media? In your answer, you should only consider the interests of the manager and workers, not those of possible third parties.

2 pts · Open

+2 points

Prohibiting social media use reduces worker's utility by V in euro terms, while the manager's profits increases by X euro per worker. Hence, social welfare increases when $X > V$. (also fine is: $X \geq V$)

Suppose the workers earn a binding minimum wage which is so high that they never want to quit this job. What will the manager decide on allowing the use of social media by the workers?

2 pts · Open

+2 points

The manager will not allow the use of social media. This way, her payoff increases by X . (<required) (optional: The workers utility decreases by V , but they won't quit. The penalty F will induce the workers to comply because the detection probability is 1 and $F > V$.)

Suppose now that wages are set by the manager so that workers are always exactly indifferent between working for the manager and working somewhere else. What will the manager decide on allowing the use of social media by the worker?

2 pts · Open

+2 points

The manager will allow it if $V > X$. Not allowing it would raise production value by X , but increase the wage by V .

Question 7

4 pts · Last updated 30 June, 08:10

A large retail company has implemented a new bonus system for their sales people. Since implementation, the company's sales have risen by 20%.

Text

Give one reason for why we cannot conclude that the new bonus system has led to an increase in sales.

2 pts · Open

+2 points

It could be that sales have increased for other reasons, e.g. a booming economy, a competitor going bankrupt, a change in customers' tastes. (the examples are optional, not required to get the points)

Describe one way in which the company could have discovered in a highly reliable way whether the new bonus system leads to an increase in sales.

2 pts · Open

+2 points

Run a randomized controlled experiment where the new bonus system is implemented in a random selection of stores and not in the others. One can reliably tell whether the bonus system leads to additional sales by comparing the sales of two groups.